Notes: Bhattacharya, Dupas, and Kanaya (RES, 2024)

Demand and Welfare Analysis in Discrete Choice Models with Social Interactions

Contents

1 Big picture

- **Question.** How should we estimate demand and welfare effects of policies in binary choice settings when individual payoffs depend on aggregate choices?
- **Core setting.** The motivating examples are policies such as subsidies for health-product adoption or school vouchers, where one person's choice can affect other people's payoffs.
- **Main distinction.** Choice data can be sufficient to predict counterfactual demand under social interactions, but the same data are generally insufficient for welfare calculations.
- **Reason for under-identification.** A single estimated social-interaction coefficient can come from different mechanisms: conformity, learning, positive health spillovers, negative deflection effects, or free-riding.
- **Why welfare differs.** These mechanisms produce the same demand response but imply different utility changes for buyers, non-buyers, eligibles, and ineligibles.
- **Main theoretical contribution.** The paper derives distribution-free bounds on welfare effects under standard index restrictions, rather than imposing a single interpretation of the social-interaction coefficient.
- **Econometric contribution.** It gives identification and estimation methods allowing village or group effects that are unobserved, correlated with observables, and potentially unbounded.
- **Empirical application.** The method is applied to experimental data on insecticide-treated mosquito-net adoption in rural Kenya.
- **Main empirical result.** Social interactions are important for demand prediction. A targeted subsidy raises take-up among both eligible and non-eligible households through spillovers.
- **Main welfare result.** Welfare effects are bounded rather than point identified. Ignoring spillovers can substantially misstate welfare gains and deadweight loss.
- **Takeaway.** In social-interaction models, demand prediction and welfare analysis are not equivalent. Welfare requires knowing, or bounding, the mechanism behind peer effects.

2 Data and measurement

2.1 Choice environment

- Individuals make a binary choice, denoted by $A \in \{0, 1\}$.
- In the empirical application, $A = 1$ means that a household buys an insecticide-treated mosquito net (ITN).
- The price of the product is experimentally varied.
- The social interaction variable is the aggregate adoption rate in the relevant group, such as a village.

Interpretation: The key feature is that a household's payoff from buying or not buying can depend on how many others buy.

2.2 Utility with social interactions

Let y be income or wealth, p be price, π be the aggregate adoption rate, and $\eta = (\eta_0, \eta_1)$ be unobserved preference heterogeneity. The indirect utility from buying and not buying is written as

$$U_1(y - p, \pi, \eta) = \delta_1 + \beta_1(y - p) + \alpha_1\pi + \eta_1, \quad (1)$$

$$U_0(y, \pi, \eta) = \delta_0 + \beta_0 y + \alpha_0\pi + \eta_0. \quad (2)$$

The observed social-interaction coefficient in the choice probability is

$$\alpha = \alpha_1 - \alpha_0. \tag{3}$$

Interpretation: Demand identifies α, the difference between spillover effects in the two utility states. Welfare requires knowing α_1 and α_0 separately.

2.3 Choice probability

The structural choice probability is

$$q_1(p, y, \pi) = \Pr\{U_1(y - p, \pi, \eta) \geq U_0(y, \pi, \eta)\}. \tag{4}$$

Under the linear-index model,

$$q_1(p, y, \pi) = F(c_0 + c_1 p + c_2 y + \alpha\pi), \tag{5}$$

where F is the distribution of $\eta_0 - \eta_1$, $c_1 = -\beta_1$, and $c_2 = \beta_1 - \beta_0$.

Interpretation: Choice data can estimate the reduced interaction coefficient α, but cannot decompose it into the welfare-relevant components α_1 and α_0.

2.4 Equilibrium adoption

Aggregate adoption satisfies a fixed-point equation:

$$\pi = \int q_1(p, y, \pi)\, dF_Y(y). \tag{6}$$

For a subsidy policy with eligible households facing price p_1 and ineligible households facing price p_0, the post-policy equilibrium satisfies

$$\pi_1 = \int [1\{y \leq \tau\} q_1(p_1, y, \pi_1) + 1\{y > \tau\} q_1(p_0, y, \pi_1)]\, dF_Y(y). \tag{7}$$

Interpretation: Social interactions make counterfactual demand an equilibrium object. A subsidy changes demand directly through price and indirectly through π.

2.5 Compensating variation

For an eligible household, compensating variation S solves

$$\max\{U_1(y + S - p_1, \pi_1, \eta), U_0(y + S, \pi_1, \eta)\} = \max\{U_1(y - p_0, \pi_0, \eta), U_0(y, \pi_0, \eta)\}. \tag{8}$$

For an ineligible household, the price does not change, so p_1 is replaced by p_0 in the left-hand side.

Interpretation: The welfare effect is a money-metric income compensation that restores the household's realized utility to its pre-policy level.

3 Models and theoretical results

3.1 Demand is identified, welfare is not

The estimated social interaction coefficient is

$$\alpha = \alpha_1 - \alpha_0. \tag{9}$$

Several mechanisms can generate the same α:

- conformity or social learning may raise the utility from adopting, increasing α_1;
- positive health externalities may affect buyers and non-buyers differently;
- mosquito deflection can lower non-buyers' utility when others adopt, changing α_0;
- free-riding can reduce the value of adopting when others adopt.

Interpretation: Because welfare depends separately on the utility of choosing 1 and choosing 0, knowing only α is not enough.

3.2 Case 1: $\alpha_1 \geq 0 \geq \alpha_0$

This case corresponds to positive spillovers for adopters and possibly negative spillovers for non-adopters. For subsidy eligibles, the paper derives the distribution of compensating variation. In simplified form, the CDF has three regions:

$$\Pr(S^{Elig} \leq a) = \begin{cases} 0, & a < p_1 - p_0 - \frac{\alpha_1}{\beta_1}(\pi_1 - \pi_0), \\ q_1\left(p_1 - a, y, \pi_0 + \frac{\alpha_1}{\alpha}(\pi_1 - \pi_0)\right), & \text{intermediate } a, \\ 1, & a \geq \frac{\alpha - \alpha_1}{\beta_0}(\pi_1 - \pi_0). \end{cases} \tag{10}$$

Interpretation: The middle term depends on α_1, which is not identified from demand. Therefore welfare is bounded rather than point identified.

3.3 Mean welfare bounds

The mean compensating variation is obtained from the CDF by

$$E[S] = \int_0^\infty [1 - F_S(a)]da - \int_{-\infty}^0 F_S(a)da. \tag{11}$$

Because $\alpha_1 \in [0, \alpha]$ in this case, one obtains lower and upper bounds by varying α_1 over the feasible interval.

Interpretation: The bounds are distribution-free in the sense that the paper does not need to assume a single structural mechanism behind the social-interaction coefficient.

3.4 Case 2: $\alpha_1 \geq \alpha_0 \geq 0$

This case corresponds to positive effects in both adoption states, with the adopting state receiving the larger spillover. The paper derives analogous expressions $C_1(\alpha_1)$ and $C_2(\alpha_1)$ and then obtains lower and upper bounds:

$$LB^{Elig}_{\alpha_1 \geq \alpha_0 \geq 0} = \min\left\{\inf C_1(\alpha_1), \inf C_2(\alpha_1)\right\}, \tag{12}$$

$$UB^{Elig}_{\alpha_1 \geq \alpha_0 \geq 0} = \max\left\{\sup C_1(\alpha_1), \sup C_2(\alpha_1)\right\}. \tag{13}$$

Combining the two cases gives wider overall bounds:

$$LB^{Elig} = \min\{LB^{Elig}_{\alpha_1 \geq \alpha_0 \geq 0}, LB^{Elig}_{\alpha_1 \geq 0 \geq \alpha_0}\}, \tag{14}$$

$$UB^{Elig} = \max\{UB^{Elig}_{\alpha_1 \geq \alpha_0 \geq 0}, UB^{Elig}_{\alpha_1 \geq 0 \geq \alpha_0}\}. \tag{15}$$

Interpretation: The paper separates restrictions that are identifiable from demand from restrictions that are welfare assumptions.

3.5 Unobserved village effects

The empirical model allows village effects ξ_v that may be correlated with observed covariates. The paper uses correlated random effects ideas and within-village transformations to identify the index parameters and welfare bounds.

Interpretation: This is important because villages can differ in malaria risk, norms, local infrastructure, and unobserved preferences.

4 Identification and estimation

4.1 Bayesian equilibrium interpretation

The paper gives conditions under which agents share common beliefs about average adoption. Under those conditions, the equilibrium choice probability can be written using constant and symmetric beliefs.

Interpretation: This justifies estimating a reduced-form choice probability with village average adoption as the interaction term.

4.2 Semiparametric identification

The model identifies the index parameters in the choice probability:

$$q_1(p, y, \pi) = F(c_0 + c_1 p + c_2 y + \alpha\pi). \tag{16}$$

The parameters c_0, c_1, c_2, α can be identified under index restrictions and support conditions, but α_1 and α_0 are not separately identified.

Interpretation: Demand can be point identified; welfare is set identified.

4.3 Estimation in the application

The empirical application estimates:

- a probit model without village effects;
- a correlated random effects probit model;
- counterfactual equilibria under a targeted subsidy;
- welfare bounds using the theoretical formulas.

Interpretation: The correlated random effects model is the preferred empirical specification because it accounts for village-level unobserved heterogeneity.

5 Tables and figures

The tables and figures below are directly cropped from the paper and grouped compactly.

5.1 Tables 1–3: data and demand estimates

Read:

- Table 1 reports 2,197 households from 11 villages. About 35% bought an ITN, the mean assigned price is 129 Kenyan shillings, and average village ITN use is 0.28.
- Table 2 shows large cross-village variation in purchase rates, prices, wealth, malaria incidence, and education.
- Table 3 shows that price significantly reduces ITN purchase, while village average adoption significantly increases purchase.
- The marginal effect of price is about -0.0033 in the correlated random effects probit.
- The marginal effect of village average use is about 0.754 and statistically significant.

Economic message: The data show both standard downward-sloping demand and strong social interactions in ITN adoption.

Summary statistics ($N = 2{,}197$)

Variable	Definition	Mean	Min	Max
Adoption	Bought ITN	0.35	0.00	1.00
Price (Kenyan Shillings)	Randomly assigned price of ITN	129	0.00	300
Wealth (Kenyan Shillings)	Household wealth	21,126	0.00	99,003
Avg use	Average ITN use in village	0.28	0.10	0.59
Avg malaria	Incidence of malaria in village	0.66	0.58	0.79
Educ-female (years)	Years of school for female adults	5.87	0.00	15.00
Child	Whether HH has child under 10	0.74	0.00	1.00

Notes: Households surveyed come from 11 different villages. At the time of the study the exchange rate was approximately 65 Kenyan Shillings to 1 USD.

Table 1: Summary statistics

Village Village	Number of households	Purchased ITN	ITN Price (KSh)	Wealth (KSh)	Malaria Incidence	Years of Education of female head	HH has child under 10
1	175	0.27	142.51	22,885	0.64	5.45	0.86
2	250	0.45	108.32	23,741	0.60	5.30	0.75
3	222	0.45	120.81	20,677	0.62	4.50	0.72
4	212	0.34	129.95	23,124	0.70	7.35	0.72
5	287	0.34	117.94	23,455	0.79	6.39	0.74
6	182	0.27	140.11	16,151	0.63	4.48	0.75
7	142	0.16	179.08	11,724	0.67	6.11	0.69
8	241	0.22	180.33	24,261	0.66	8.13	0.76
9	214	0.12	156.82	22,265	0.58	5.78	0.74
10	167	0.68	61.98	17,425	0.66	4.53	0.65
11	105	0.73	49.52	20,235	0.58	5.43	0.75

Notes: Column 2 indicates the number of households sampled. We do not know the exact number of households in each village since we did not complete a census before sampling, but since households were sampled from school registers and schooling rates are high, we estimate that sampled households represent roughly 80% of the village population.

Table 2: Village averages

	Probit			Correlated random effect		
Variable	Marginal effect	Std. Error	*P*-value	Marginal effect	Std. Error	*P*-value
Price (KSh)	−0.0032	0.0002	<0.001	−0.0033	0.0002	<0.001
Wealth ('000 KSh)	0.0009	0.0006	0.132	0.0008	0.0006	0.166
Educ-female (years)	0.0037	0.0028	0.189	0.0028	0.0029	0.342
Child under 10	−0.0701	0.0251	0.005	−0.0706	0.0252	0.005
Village level Avg Use	0.6558	0.0916	<0.001	0.7539	0.1181	<0.001
Village level Avg Malaria	−0.0534	0.1746	0.760	−0.1399	0.1975	0.479

Notes: 2,197 observations from 11 villages.

Table 3: Marginal effects

5.2 Figure 1 and Table 4: demand effects of the subsidy

Read:

- Figure 1 plots predicted adoption against the fraction eligible for a subsidy.
- Ignoring spillovers over-predicts adoption at low eligibility thresholds and under-predicts adoption at high eligibility thresholds.
- Table 4 evaluates a targeted subsidy that gives eligible households the ITN for 50 KSh while non-eligible households face 250 KSh.
- Eligible demand rises from 0.012 pre-subsidy to 0.465 post-subsidy.
- The own-price effect is 0.354 and the spillover effect is 0.111 for eligibles.
- Non-eligible demand also rises, from 0.015 to 0.029, almost entirely due to spillovers.

Economic message: Spillovers matter both for those treated by the subsidy and for those who do not receive the subsidy.

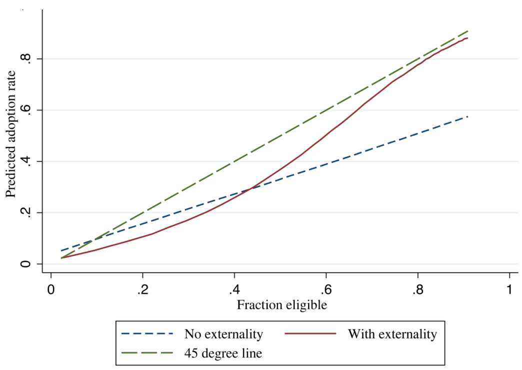

FIGURE 1

redicted equilibrium adoption of ITN under changing eligibility rule for subsidy, plotted against fraction eligible

: We consider a hypothetical subsidy in which eligible gets a price of 50Ksh for an ITN while the rest face a price of 250K lot the predicted aggregate take-up of ITNs corresponding to different eligibility shares, based on coefficients obtained from Prc ssions either including (solid) or excluding (small dash) the spillover effect. The 45° line (large dash) is shown for comparison.

Figure 1: predicted adoption under changing eligibility

Impact of subsidy on demand by eligibility

		Eligible	Non-eligible
Pre-subsidy		0.012	0.015
Post-subsidy	Overall	0.465	0.029
	Own effect	0.354	0.015
	Spillover effect	0.111	0.014

Notes: The table shows estimated demand. See Section 7 for details.

Table 4: demand effect by eligibility

5.3 Table 5: welfare gains and deadweight loss

Read:

- Without externality, the probit model gives an eligible welfare gain of 52.58 KSh and a net welfare gain of 13.43 KSh.
- With externality, welfare becomes bounded because the interaction coefficient cannot be decomposed into its underlying mechanisms.
- In the correlated random effects model with $\alpha_1 \geq 0 \geq \alpha_0$, eligible welfare lies between 1.52 and 37.72 KSh, while ineligible welfare lies between -36.25 and 0.60 KSh.
- Under broader externality assumptions, the welfare bounds become much wider.
- Deadweight loss can be positive or negative depending on how spillovers map into welfare.

Economic message: Choice data alone do not pin down whether a subsidy is welfare improving once social interactions are present. Bounds depend on explicit assumptions about spillover mechanisms.

TABLE 5

Mean welfare gain and deadweight loss in Kenyan shillings

	W/O externality				With externality								
	Welfare eligible	Cost	Net welfare gain	Deadweight loss	Welfare eligible		Welfare ineligible		Cost	Net welfare gain		Deadweight loss	
					LB	UB	LB	UB		LB	UB	LB	UB
Probit	52.58	23.70	13.43	10.27	12.94	43.29	−25.13	0.73	23.70	−15.41	11.60	12.10	39.1
Probit with correlated random effects ($\sigma_e = 0.2$)													
if $\alpha_1 \geq 0 \geq \alpha_0$	23.89	23.70	6.10	17.60	1.52	37.72	−36.25	0.60	23.70	−26.60	10.08	13.62	50.3
bounds if $600\alpha \geq \alpha_1 \geq \alpha_0 \geq 0$				9.51	2, 343.56	0.00	75.66	23.70	2.43	654.76	−631.07	21.27	
bounds if $\alpha_1 \geq 0 \geq \alpha_0$ or $600\alpha \geq \alpha_1 \geq \alpha_0 \geq 0$				1.52	2, 343.56	−36.25	75.66	23.70	−26.60	654.76	−631.07	50.30	

Notes: The table shows estimated welfare effects. The share eligible in the sample is 0.26. LB, lower bound; UB, upper bound.

5.4 Figure 2: equilibrium uniqueness in the empirical application

Read:

- Figure 2 plots the empirical fixed-point objective function for each of the 11 villages.
- The objective function is convex with a clear minimum in each village.
- The minima are generally near 0.15, except in villages 7 and 10, where predicted adoption is higher.
- The figure supports the empirical use of a unique equilibrium in the welfare calculations.

Economic message: The theory allows for equilibrium complications, but the empirical application detects a unique solution in the counterfactual calculations.

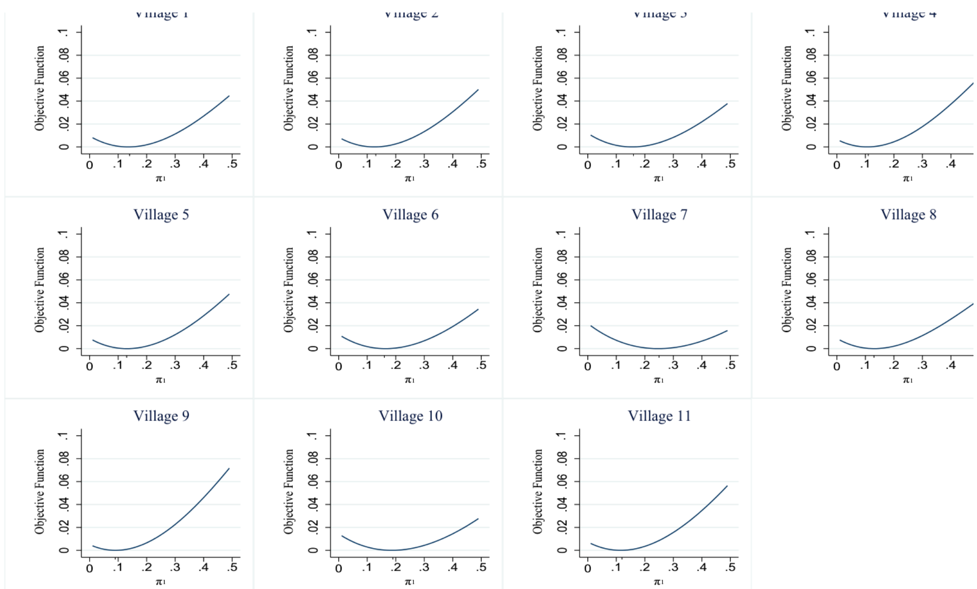

FIGURE 2

Objective function for each of the 11 villages

6 Short summary

- The paper studies demand and welfare in binary choice models with social interactions.
- It shows that demand prediction and welfare calculation differ under social spillovers.
- Choice data identify the aggregate social interaction coefficient, not the separate utility effects behind it.

- Distinct mechanisms such as conformity, learning, positive health effects, deflection, and free-riding can generate the same demand coefficient.
- The paper derives welfare bounds under standard index restrictions.
- It develops methods allowing unobserved group effects that can be correlated with observables.
- The empirical application uses randomized ITN subsidy data from rural Kenya.
- Demand is downward sloping in price and positively affected by village adoption.
- Subsidies have direct own-price effects and indirect spillover effects.
- Welfare effects are generally bounded, not point identified.
- Ignoring spillovers can substantially misstate adoption, welfare gains, and deadweight loss.

7 Conclusion

7.1 Core conclusion

Bhattacharya, Dupas, and Kanaya show that in discrete choice models with social interactions, counterfactual demand can be estimated without enough information to conduct welfare analysis. Welfare depends on the mechanism generating the social interaction coefficient, not just on the coefficient itself.

7.2 Economic intuition

A subsidy changes demand through price and through equilibrium spillovers. But whether spillovers raise or lower welfare depends on why people care about others' choices. If others' adoption creates learning or conformity benefits, the welfare implications differ from a case in which others' adoption deflects mosquitoes toward non-adopters or enables free-riding. Demand sees only the net interaction; welfare needs the mechanism.

7.3 Takeaway

The paper's main lesson is methodological: applied researchers using social-interaction demand models should not move directly from counterfactual demand to welfare. They should either justify the underlying spillover mechanism or report welfare bounds, as this paper proposes.